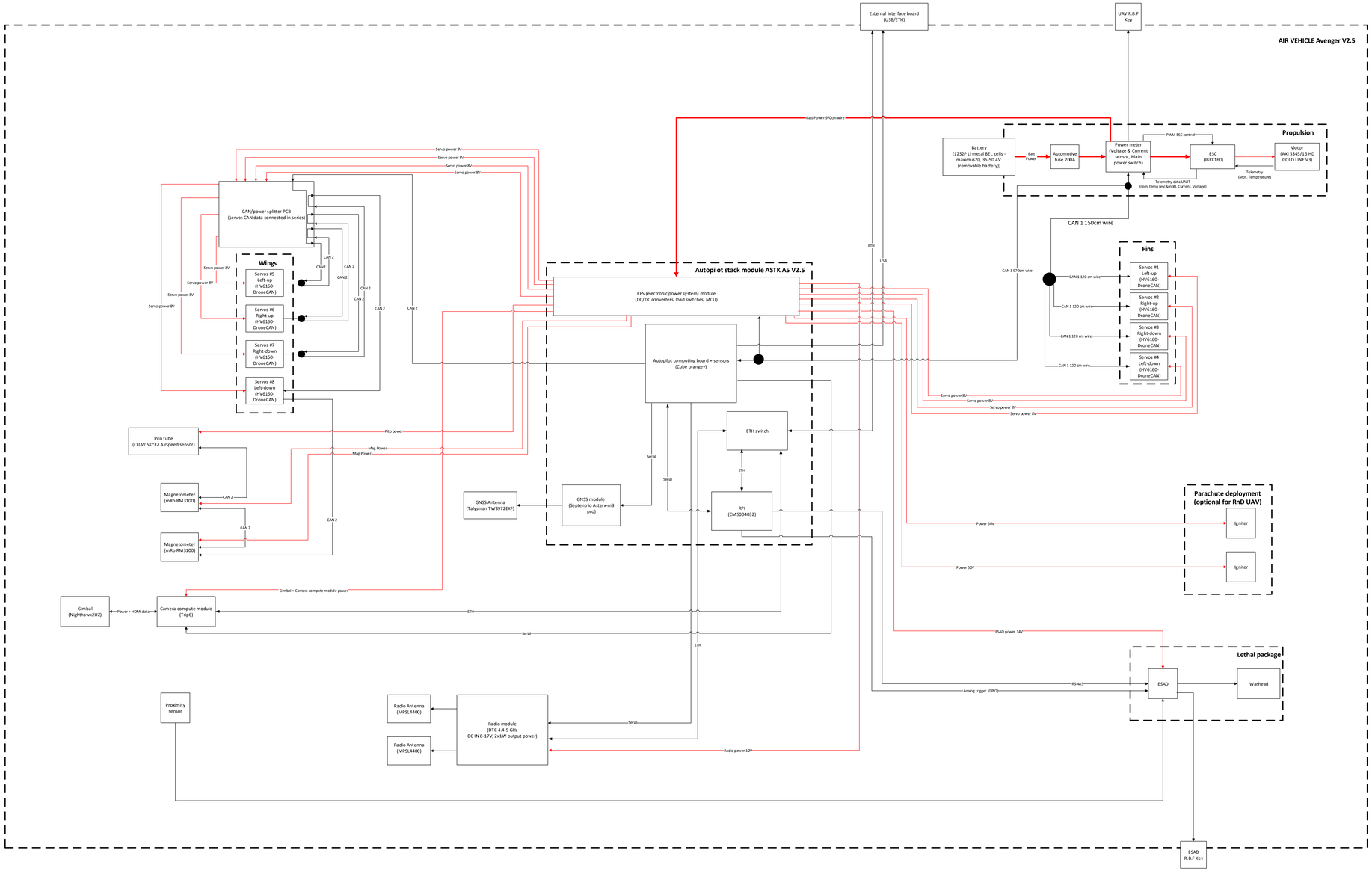


Main component functions

Autopilot stack module ASTK A5 V2.5

Autopilot computing board + sensors: Responsible for main UAV control logic, flight stabilization and navigation. Provides attitude, position and kinematics estimation from fusing internal end external sensor data.

EPS (electronic power system) module: Responsible for providing different power levels to the UAV's subsystems

ETH switch: Responsible for interfacing different aircraft system which use IP network for communication.

RPI: Responsible for interfacing between aircraft systems and SC2 command and control software.

GNSS module: Responsible for providing absolute position and velocity estimates from received GNSS signal data.

GNSS Antenna: Responsible for receiving GNSS satellite signals.

Propulsion

Battery: Responsible for storing and supplying the electrical power required for UAV flight.

Automotive Fuse (200 A): Provides protection in the event of a short circuit in the main power line.

Power Meter: Responsible for controlling the UAV's ON/OFF state. When the UAV R.B.F. key is removed, the UAV is switched ON. Provides battery voltage and current metrics to the autopilot.

ESC (Electronic Speed Controller): Responsible for controlling motor speed. It converts the battery's DC power into the appropriate three-phase electrical signals required to drive the motors.

Motor: Converts electrical power into mechanical power to generate thrust.

Fins

Servo #1-#4: Responsible for UAV pitch and yaw control. The UAV does not bank during turns like conventional UAVs; instead, it performs skid turns using yaw. All four fins are mixed to provide both pitch and yaw control.

Wings

Servo #5-#8: Responsible for UAV roll stabilization. The UAV continuously stabilizes its roll to maintain a level attitude relative to the horizon. During yaw turns, the UAV does not roll; instead, it maintains a level attitude and performs a horizontal skid turn.

Pito tube

Responsible for measuring airspeed. Has in-built heating and drainage.

Magnetometer

Responsible for measuring the UAV's heading. The current magnetometer modules also include barometric sensors, which are used for altitude estimation.

Gimbal

Responsible for IR/RGB camera sensor pointing and stabilization.

Camera compute module

Responsible for controlling the gimbal, providing video feed, target tracking, filtered gimbal kinematic measurements, and target position estimation based on provided UAV position data.

Radio module

Responsible for receiving command and control data, while transmitting telemetry and video.

Radio Antenna

Responsible for transmitting and receiving radio signals to and from the radio module.

Lethal package

ESAD (Electronic Safe and Arm Device): is a high-voltage, all-electronic safety device used to control in-line the initiation sequence of a pyrotechnic train on warhead modules. Essentially, it is a safety device used to prevent the accidental arming or firing of a pyrotechnic event by eliminating system sensitivities to environmental stimuli, such as electromagnetic fields, electrostatic discharge, and stray voltages.

ESAD R.B.F Key: Responsible for ensuring the safe arming sequence of the ESAD. It must be removed according to the specific arming sequence of the ESAD

Proximity sensor: Responsible for sending the triggering signal to the ESAD

Warhead: Responsible for explosion

Parachute deployment

Igniters: Pyrotechnic devices responsible for burning the wire that holds the payload bay cover, where the parachute is installed

External interface board

Responsible for connecting to the UAV's subsystems.